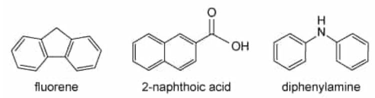
A student separated a 2-g mixture of fluorene, 2-naphthoic acid, and diphenylamine (shown below) dissolved in 25 mL of the organic solvent diethyl ether.



The separation experiment was performed according to the following steps:

Step 1

The mixture solution was partitioned by pouring it into a separatory funnel containing 10 mL water, and the organic layer was extracted with 10 mL 1 M NaOH(*aq*).

Step 2

The aqueous layer was transferred to Flask A, and concentrated HCl(*aq*) was added dropwise to the flask until pH paper indicated that the solution was acidic. Flask A was then placed in ice water to induce precipitate formation.

Step 3

The organic layer was extracted with 10 mL 3 M HCl(*aq*). The resulting aqueous layer was then transferred to Flask B and 8 M NaOH was added dropwise to Flask B until pH paper indicated that the solution was basic. Flask B was then placed in ice water, to maximize precipitate formation.

Step 4

The precipitates in flasks A and B were collected separately by vacuum filtration, rinsed with distilled water, and dried in an oven.

Step 5

The organic layer was transferred into Flask C and dried by adding anhydrous $MgSO_4$ to remove any residual water present. The organic solvent was separated from the $MgSO_4$ in Flask C by gravity filtration, and the solvent was evaporated, yielding white crystals.

Step 6

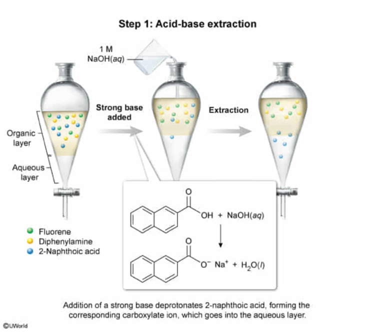
The purity of the three isolated compounds was analyzed by thin-layer chromatography (TLC) on silica plates. A sample of the mixture was spotted on the TLC plate as a reference in addition to the isolated compound.

In which of the following steps is 2-naphthoic acid separated from the mixture?

- ✔ A. Step 1 [36%]
- B. Step 3 [40%]
- C. Step 4 [16%]
- D. Step 5 [6%]

Correct
36% answered correctly

Explanation:



A mixture of organic molecules with acidic and basic functional groups can be separated by and **acid-base** extraction. The acid and base components in the mixture are initially in the organic layer but can enter the aqueous phase if converted to an ionic salt by deprotonation or protonation via extraction with an aqueous base or acid, respectively.

Carboxylic acids can be deprotonated by strong and weak bases. Consequently, separating a mixture of a carboxylic acid and an organic base can be achieved by extraction with an aqueous base that can deprotonate the carboxylic acid and pull it into the aqueous layer. The organic base in the mixture is not reactive toward the aqueous base added for the extraction and remains in the organic layer.

The passage states that in Step 1, the mixture of fluorene, 2-naphthoic acid, and diphenylamine is extracted with 1 M NaOH(*aq*). 2-

naphthoic acid is a carboxylic acid and is deprotonated by NaOH(*aq*) during the extraction, forming a carboxylate salt that is pulled into the aqueous layer. Because fluorene is not sufficiently acidic (pK_a = 22) and diphenylamine is a base, these molecules are not deprotonated by NaOH(*aq*) and remain in the organic layer. Therefore, Step 1 separates 2-naphthoic acid from the mixture.

(Choice B) Step 3 describes the separation of fluorene and diphenylamine by extraction with HCl(*aq*), which protonates diphenylamine to form an ammonium salt that goes into the aqueous layer.

(Choice C) Step 4 describes the isolation of 2-naphthoic acid and diphenylamine (ie, the precipitates in Flask A and B, respectively) by vacuum filtration.

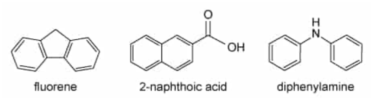
(Choice D) Step 5 describes isolation of fluorene from the organic layer.

Educational objective:
A mixture of organic molecules with acidic or basic functional groups can be separated by an acid-base extraction. The acidic or basic molecule can enter the aqueous layer when extracted with a base or acid, respectively.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403146
System:
Separation Techniques, Spectroscopy, and Analytical Methods

A student separated a 2-g mixture of fluorene, 2-naphthoic acid, and diphenylamine (shown below) dissolved in 25 mL of the organic solvent diethyl ether.



The separation experiment was performed according to the following steps:

Step 1

The mixture solution was partitioned by pouring it into a separatory funnel containing 10 mL water, and the organic layer was extracted with 10 mL 1 M NaOH(*aq*).

Step 2

The aqueous layer was transferred to Flask A, and concentrated HCl(*aq*) was added dropwise to the flask until pH paper indicated that the solution was acidic. Flask A was then placed in ice water to induce precipitate formation.

Step 3

The organic layer was extracted with 10 mL 3 M HCl(*aq*). The resulting aqueous layer was then transferred to Flask B and 8 M NaOH was added dropwise to Flask B until pH paper indicated that the solution was basic. Flask B was then placed in ice water, to maximize precipitate formation.

Step 4

The precipitates in flasks A and B were collected separately by vacuum filtration, rinsed with distilled water, and dried in an oven.

Step 5

The organic layer was transferred into Flask C and dried by adding anhydrous $MgSO_4$ to remove any residual water present. The organic solvent was separated from the $MgSO_4$ in Flask C by gravity filtration, and the solvent was evaporated, yielding white crystals.

Step 6

The purity of the three isolated compounds was analyzed by thin-layer chromatography (TLC) on silica plates. A sample of the mixture was spotted on the TLC plate as a reference in addition to the isolated compound.

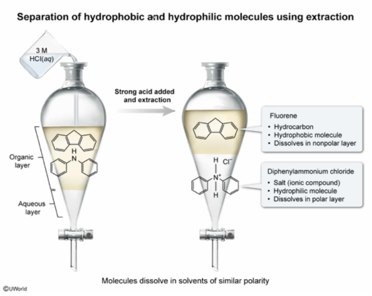
Why does fluorene remain in the organic layer during the extraction with 3 M HCl(*aq*)?

I. Fluorene cannot accept H^+ from HCl(*aq*).
II. Fluorene is hydrophilic and remains in the nonpolar layer.
III. Fluorene is hydrophobic and remains in the nonpolar layer.

- A. I only [3%]
- B. II only [2%]
- C. III only [17%]
- D. I and III only [77%]

Correct
76% answered correctly

Explanation:



Extraction is a technique used to separate molecules based on their **solubility**. Molecules dissolve in solvents of **similar polarity** (ie, nonpolar molecules dissolve in nonpolar solvents and polar molecules dissolve in polar solvents).

Fluorene is a hydrocarbon (ie, composed of only carbon and hydrogen), which cannot act as a Brønsted-Lowry base to accept a proton (H^+) from an acid such as HCl **(Number I)**. Hydrocarbons do not contain electronegative atoms (O, N, or F) and therefore are hydrophobic (nonpolar) molecules. Because molecules dissolve in solvents of similar polarity, fluorene remains in the organic solvent, which is nonpolar relative to the aqueous layer **(Number III)**.

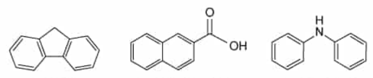
(Number II) Because fluorene does not contain electronegative atoms such as O, N, or F, it does not exhibit strong intermolecular interactions with water (eg, hydrogen bonding) and cannot dissolve in water. Therefore, fluorene is considered *hydrophobic* rather than hydrophilic.

Educational objective:
Extraction is a technique used to separate molecules based on their solubility. Molecules will dissolve in solvents of similar polarity.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403147
System:
Separation Techniques, Spectroscopy, and Analytical Methods

A student separated a 2-g mixture of fluorene, 2-naphthoic acid, and diphenylamine (shown below) dissolved in 25 mL of the organic solvent diethyl ether.



fluorene 2-naphthoic acid diphenylamine

The separation experiment was performed according to the following steps:

Step 1

The mixture solution was partitioned by pouring it into a separatory funnel containing 10 mL water, and the organic layer was extracted with 10 mL 1 M NaOH(*aq*).

Step 2

The aqueous layer was transferred to Flask A, and concentrated HCl(*aq*) was added dropwise to the flask until pH paper indicated that the solution was acidic. Flask A was then placed in ice water to induce precipitate formation.

Step 3

The organic layer was extracted with 10 mL 3 M HCl(*aq*). The resulting aqueous layer was then transferred to Flask B and 8 M NaOH was added dropwise to Flask B until pH paper indicated that the solution was basic. Flask B was then placed in ice water, to maximize precipitate formation.

Step 4

The precipitates in flasks A and B were collected separately by vacuum filtration, rinsed with distilled water, and dried in an oven.

Step 5

The organic layer was transferred into Flask C and dried by adding anhydrous $MgSO_4$ to remove any residual water present. The organic solvent was separated from the $MgSO_4$ in Flask C by gravity filtration, and the solvent was evaporated, yielding white crystals.

Step 6

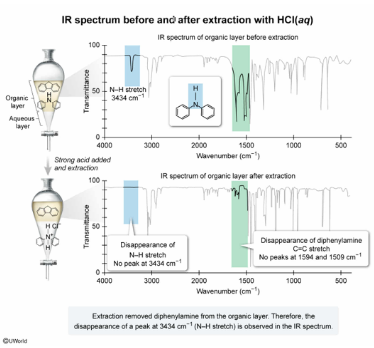
The purity of the three isolated compounds was analyzed by thin-layer chromatography (TLC) on silica plates. A sample of the mixture was spotted on the TLC plate as a reference in addition to the isolated compound.

The IR spectrum of the organic layer is obtained before and after extraction with HCl(*aq*), and the disappearance of an N–H stretch is observed. Compared to the IR spectrum before the extraction, the IR spectrum after the extraction is expected to be missing a peak at which wavenumber as a result of the lost N–H stretch?

- A. 1480 cm^{-1} [10%]
- B. 1710 cm^{-1} [16%]
- C. 3050 cm^{-1} [27%]
- D. 3434 cm^{-1} [45%]

Correct
44% answered correctly

Explanation:



In infrared (IR) spectroscopy, the functional groups in a molecule absorb IR radiation at different frequencies and different intensities depending on the types of atoms and bonds present. Therefore, the signal for each functional group appears in a particular region of the IR spectrum (ie, a plot of transmittance vs. wavenumber). IR spectroscopy can be used to **detect** the structural changes in a molecule during a chemical reaction or **changes in a mixture** before and after separation of some mixture components by **observing** the

appearance or disappearance of peaks that are unique to a given molecule.

Step 3 of the passage describes an extraction with HCl(*aq*), in which diphenylamine is protonated, forming a water-soluble ammonium salt. Because the ammonium salt of diphenylamine goes into the aqueous layer during the extraction, stretches unique to diphenylamine, such as an N–H stretch, will be absent in the IR spectrum of the organic layer after the extraction. **N–H stretches** absorb infrared light between **3600 - 3200 cm^{-1}**. Therefore, the disappearance of a peak at 3434 cm^{-1} corresponds to an N–H stretch.

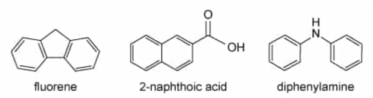
(Choices A and C) Peaks at 1480 cm^{-1} and 3050 cm^{-1} correspond to a C=C stretch and an sp^2 C–H stretch, respectively. These peaks are present in the IR spectra before and after the extraction because *both* fluorene and diphenylamine contain C=C and sp^2 C–H bonds.

(Choice B) A peak at 1710 cm^{-1} corresponds to a C=O stretch. A peak at this frequency should not be observed in either IR spectrum because neither diphenylamine nor fluorene contain a C=O bond.

Educational objective:
Functional groups in a molecule absorb at different frequencies and intensities in an infrared (IR) spectrum. N–H stretches have a characteristic absorption between 3600 - 3200 cm^{-1}. IR spectroscopy can be used to detect structural changes to a molecule during a chemical reaction or changes in a mixture due to separation of the mixture components by observing the appearance or disappearance of peaks that are unique to a given molecule.

Time spent:0 sec	**QID**:403148
Subject:	**System:**
Organic Chemistry	Separation Techniques, Spectroscopy, and Analytical Methods

A student separated a 2-g mixture of fluorene, 2-naphthoic acid, and diphenylamine (shown below) dissolved in 25 mL of the organic solvent diethyl ether.



The separation experiment was performed according to the following steps:

Step 1

The mixture solution was partitioned by pouring it into a separatory funnel containing 10 mL water, and the organic layer was extracted with 10 mL 1 M NaOH(*aq*).

Step 2

The aqueous layer was transferred to Flask A, and concentrated HCl(*aq*) was added dropwise to the flask until pH paper indicated that the solution was acidic. Flask A was then placed in ice water to induce precipitate formation.

Step 3

The organic layer was extracted with 10 mL 3 M HCl(*aq*). The resulting aqueous layer was then transferred to Flask B and 8 M NaOH was added dropwise to Flask B until pH paper indicated that the solution was basic. Flask B was then placed in ice water, to maximize precipitate formation.

Step 4

The precipitates in flasks A and B were collected separately by vacuum filtration, rinsed with distilled water, and dried in an oven.

Step 5

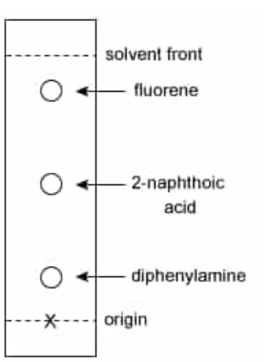
The organic layer was transferred into Flask C and dried by adding anhydrous $MgSO_4$ to remove any residual water present. The organic solvent was separated from the $MgSO_4$ in Flask C by gravity filtration, and the solvent was evaporated, yielding white crystals.

Step 6

The purity of the three isolated compounds was analyzed by thin-layer chromatography (TLC) on silica plates. A sample of the mixture was spotted on the TLC plate as a reference in addition to the isolated compound.

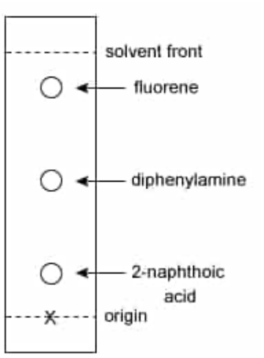
Which of the following TLC plates correctly represents the relative positions of the mixture components?

○ A.



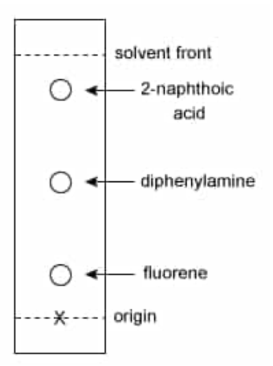
[4%]

✔ ● B.



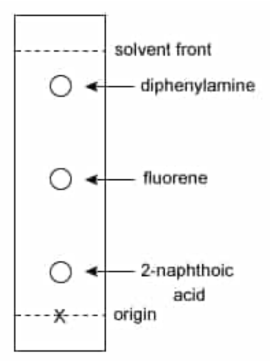
[73%]

◯ C.



[19%]

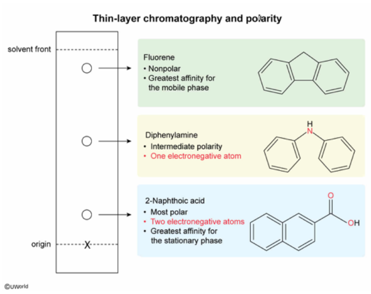
◯ D.



[2%]

Correct
74% answered correctly

Explanation:



Thin-layer chromatography (TLC) is a technique used to **separate** the **components of a mixture** based on polarity. In normal-phase TLC, nonpolar compounds have stronger intermolecular interactions with the nonpolar mobile phase, less affinity for the polar stationary phase than polar compounds and therefore travel farther up the TLC plate.

Comparing the structures of the mixture components:

- Fluorene is a nonpolar hydrocarbon.
- Diphenylamine contains nonpolar aromatic rings and a polar secondary amine.
- 2-naphthoic acid contains a nonpolar naphthalene ring and a polar carboxyl (–CO_2H) group.

Because fluorene is nonpolar, it has a strong affinity for the mobile phase and travels farthest up the TLC plate. The amine and carboxyl group in diphenylamine and 2-naphthoic acid, respectively are both hydrogen bond donors and acceptors and interact with the stationary phase through strong intermolecular interactions (hydrogen bonds). However, diphenylamine has one electronegative atom whereas 2-naphthoic acid has two electronegative atoms. Therefore, 2-naphthoic acid is more polar and has a stronger affinity for the stationary phase than diphenylamine.

Therefore, fluorene is the least polar and appears closest to the solvent front. Diphenylamine has intermediate polarity, and 2-naphthoic acid is the most polar and appears closest to the origin. Only **Choice B** shows the mixture components separated in this order.

(Choice A) This TLC plate shows 2-naphthoic acid traveled farther than diphenylamine. Because 2-naphthoic acid is more polar than diphenylamine, 2-naphthoic acid travels a *shorter* distance up the TLC plate.

(Choice C) This TLC plate shows the order of the mixture components if reverse-phase chromatography were used.

(Choice D) This TLC plate shows diphenylamine traveled farther than fluorene. Because diphenylamine is more polar than fluorene, diphenylamine travels a shorter distance up the TLC plate than fluorene.

Educational objective:
TLC is a technique used to separate the components of a mixture based on polarity. The ratio of the distance traveled by a compound up the TLC plate to the distance traveled by the mobile phase is known as the R_f value.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403149
System:
Separation Techniques, Spectroscopy, and Analytical Methods